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CHE 321; Spring - 2020; Quiz - 1

100 Points

1. (25 points) What is the meaning of 'order of reaction'?

25 The order of reaction Chemically is the power raised to the concentration $\rightarrow -r_A = k C_A^\alpha C_B^\beta \rightarrow A$ is in the order of α , B is in the order of $\beta \rightarrow$ Total order of reaction is $(\alpha + \beta)$. The order of the reaction depends on how many concentrations affect the rate of the reaction.

2. (25 points) For a single liquid-phase reaction, which reactor will lead to a higher conversion: A plug-flow reactor (PFR) or a continuous stirred tank reactor (CSTRs)?

25 PFR

CSTR

3. (25 points) A catalytic CSTR reactor of volume
- V
- (
- m^3
-) has
- W
- (Mass) Kgs of the catalyst. What is the relationship between
- k_A
- and
- k'_A
- for this reactor? Note, the bulk density of the catalyst is
- $\rho_B = W/V$
- .

$$-r_A = k C_A$$

$$-r'_A = k' C_A$$

$$F_A = C_A V$$

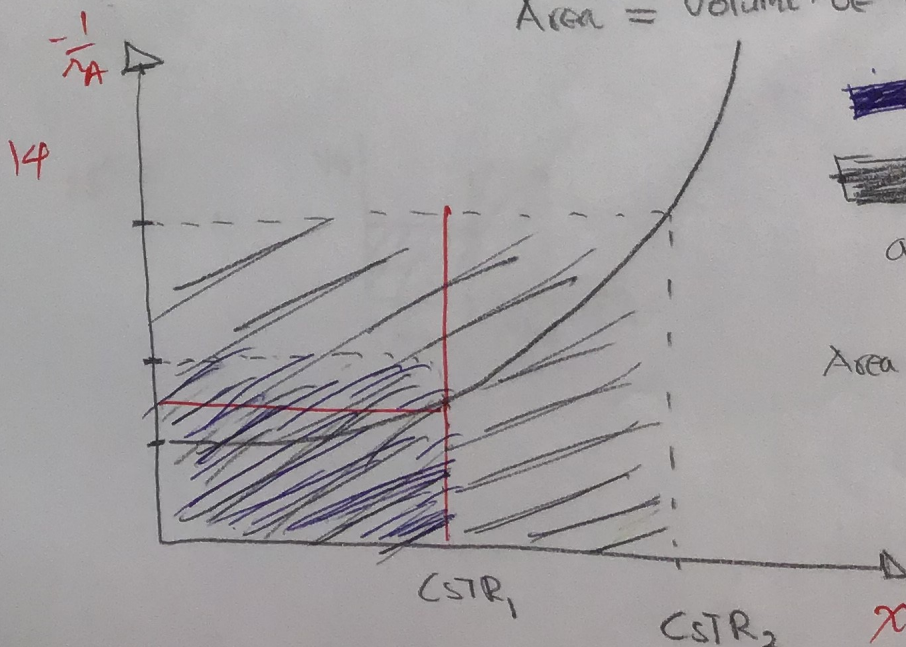
$$r_A = r'_A \rho_B$$

$$25 \quad k C_A = k' C_A \rho_B \rightarrow$$

$$k = k' \rho_B$$

4. (25 points) Qualitatively, draw a Levenspiel's plot for two CSTRs in series.

Area = Volume of reactor.



Area for CSTR(1)

Area for CSTR(2)

and it is the total Area for b

Area of CSTR(1) + CSTR(2)

is the total Area (Volume)